1. Experiment Overview

Reproduce the baseline beam prediction model from the reference paper using position-only GPS features (5D). Architecture: CNN encoder + GRU encoder-decoder + FC classifier, trained on DeepSense 6G Scenario 23.

Training Configuration

Seed 42
Batch Size 8
Epochs 20
Learning Rate 5e-4
LR Schedule [12, 18]
Device cuda

Dataset

Train 7,387 (64.9%)
Val 1,694 (14.9%)
Test 2,306 (20.3%)

2. Results - Accuracy by Prediction Step

Step	Top-1 (%)	Top-3 (%)	Top-5 (%)	PL (dB)
v=0	75.63	97.13	98.94	0.197
v=1	75.83	97.63	99.09	0.171
v=2	75.23	97.68	99.09	0.180
v=3	72.26	97.23	98.99	0.216
Overall	**74.74**	**97.42**	**99.03**	**0.191**

3. Accuracy by Speed Category

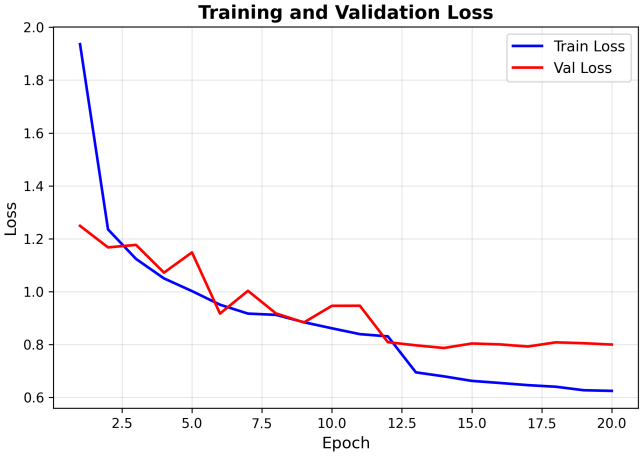
Category	Top-1 (%)	Top-3 (%)	Samples
Slow (<=10 mph)	**80.15**	**98.86**	**1363**
Medium (10-20 mph)	65.18	97.00	308
Fast (>20 mph)	60.63	91.59	315

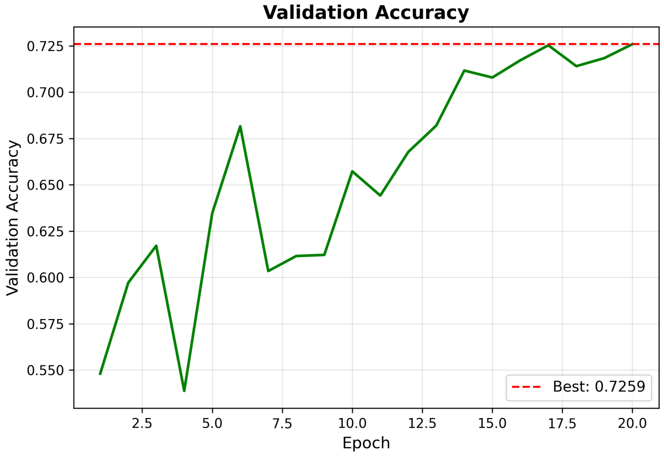
4. Comparison with Reference Paper

Metric	Paper	Ours	Diff	Status
Top-1 Acc	70.5%	74.74%	+4.24%	FAIL
Power Loss	0.59 dB	0.191 dB	-0.399 dB	FAIL

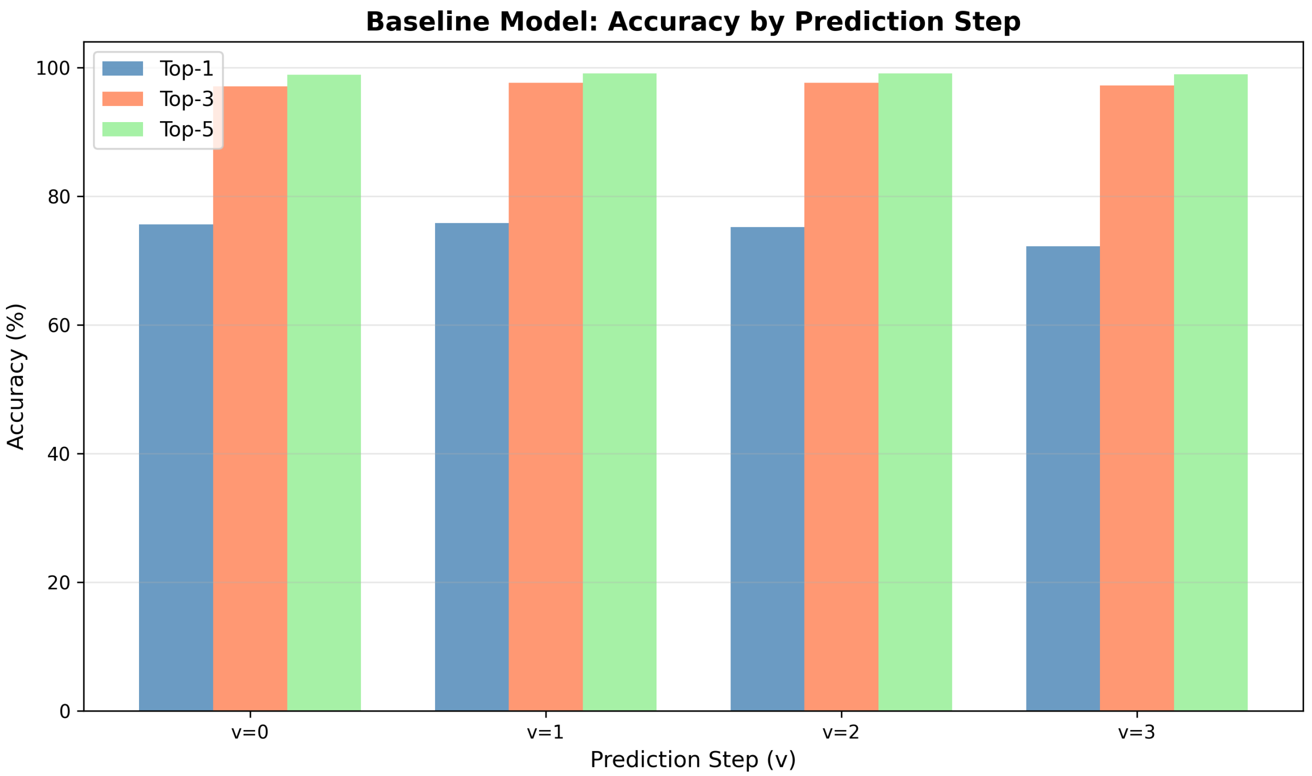
FAIL Some metrics outside tolerance - see details above.

5. Training Curves





6. Accuracy by Prediction Step



7. Training Summary

Best Epoch 19

Best Val Accuracy 0.7259

Final Train Loss 0.6248

Final Val Loss 0.8000